Validating the Use of Yerkes Observatory Glass Plates in the Long Baseline Study of Period-Changing Mira Variable Stars

Karamias Antoine[1,2]

[1]Yerkes Observatory
[2]Washington University in St Louis

Abstract

We investigate the possibility of utilizing Yerkes photographic glass plates to conduct long baseline studies of Long Period Variable stars (LPVs), and more specifically, Mira variable stars. The Yerkes Observatory Glass Plate collection contains more than 180,000 glass plates with data dating back to the late 1800s. Glass plates containing Miras will address the gaps in the timeline of observations to better constrain the light curves of Miras, allowing for more precise studies of potential changes in period. By combining glass plate data, modern databases, and new observations of the Mira variable R Aquilae, we demonstrate the potential to conduct further studies of Mira variables with fewer recorded observations to construct light curves and uncover new period changing variables.

1 Introduction

1.1 Overview of Miras

The target of this project is Long Period Variable Stars. LPVs have four categories based on amplitude and lightcurve regularity, namely Miras, semiregulars, OSARGs, and Type II Cepheids. This project will focus on Mira Variable stars, which are named for their prototype star, Mira. As with most LPVs, Miras are cool, luminous AGB stars, but they are specifically characterised by a large amplitude, a regular lightcurve, and their oscillation in the fundamental mode. They have periods anywhere from 100-1000 days long, and their visual amplitudes range from about 2.5 to around 11 magnitudes. Miras may be classified by their spectra, which indicate different evolutionary stages. M-type have more oxygen than carbon, C-type have more carbon than oxygen, and S-type have nearly equal amounts of carbon and oxygen yet more ZrO, YO, and ScO molecular absorption (Martin et al. 2016 [1], Merrill et al. 1962 [3]).

Within Miras, there are three types of period changes, first classified by Zijlstra and Bedding (2002) [5]. Continuous Period Changes (CPC) are long and continuous changes in period with no epochs or stable periods. The next type of period change is Meandering Period Change (MPC), when the period meanders around by up to 10 percent of the average period before returning. Though it progresses at a similar rate to other period changes, its overall period change is lower; however, this is the most common period change, with 10 percent of Miras undergoing MPCs in their lifetime, as compared to about 1-2 percent for the other categories (Merchán Benítez et al. 2023 [2]). Finally, there are Sudden Period Changes (SPC), which are quick changes after a long phase of stable pulsation. Miras undergoing SPC reach similar amounts of variation as the other categories in a much shorter timeframe.

Though the majority of Mira periods have remained stable, changing periods have been detected since 1841, when R Hya was determined to have a significant period change (Zijlstra and Bedding 2002 [5]). The star, first discovered by Johannes Hevelius in 1662, was theorized to undergo period changes due to thermal pulses by Wood and Zarro in 1981 (Zijlstra and Bedding 2002 [5]). Since then other theories have been put forth to explain period changes, but thermal pulses remain the prominent cause for certain types of period changes.

Thermal pulses or helium shell flashes typically occur when a star is on the asymptotic giant branch. At this point, the star is in the giant phase and it has burned through most of the helium in its core. It is left with a carbon and oxygen core and a thin shell where helium fusion continues. When the star is depleted of helium, hydrogen fusion restarts in the next layer. Once more helium is accumulated from the hydrogen fusion, helium fusion reignites, causing the star to grow and brighten for a short time (Zijlstra and Bedding 2002 [5]). Pulses can last for a few hundred years and occur every 10,000 to 100,000 years or so, and they may cause mass loss. Templeton and Mattei (2005) [4] completed an in-depth study of eight Miras with significant period changes, six of which they determined could be in some stage of a thermal pulse, bringing forth the question of what other phenomena could be behind changing periods.

Templeton and Mattei (2005) [4] offer non-evolutionary models for period changes, and Merchán Benítez et al. (2023) [2] expand upon evolutionary models. Merchán Benítez et al. (2023) [2] hold that thermal pulses may be behind CPC and SPC, but suggest alternative causes for MPC. Theories suggested by both groups include thermal relaxation oscillations after a thermal pulse, low-dimensional chaotic behavior, and entropy structure modifications due to mode-switching and regaining equilibrium (Templeton and Mattei 2005 [4]). Third Dredge Up events (3DUP) are also considered as a way to link asymmetric light curves to mass-lass rates within Miras (Merchán Benítez et al. 2023 [2]).

1.2 Significance of Yerkes

When it comes to LPVs, it is particularly important to have a long baseline of observations available for analysis. While databases from amateur astronomer organizations, such as AAVSO, tend to extend far back, there are often gaps in the data and occasionally no data on a particular star at all (see the discussion of ST Sct in section 1.3). In Figure 1, you can see the gaps glass plates could fill in for the Mira variable R Aquilae.

Though AAVSO and other organizations have collected extensive observations on R Aql, the study of R Aql will serve as validation for the methodology of glass plate analysis. By utilizing a star that has been extinsively studied to show a CPC, we can demonstrate the significance of including Yerkes glass plates in future Mira investigations.

Additionally, photographic plates can have around 0.05-0.1 magnitude precision per epoch, which provides a factor of 2-4 times improvement to what typical AAVSO visual estimates can deliver. Although the Harvard DASCH collection has set the precedent for the feasibility of glass plate data analysis, Yerkes offers an independent archive, equipped with different observers, sites, emulsions, and exposure times. Different telescopes and cameras even provide varying field-of-view density, all of which contribute to the uniqueness of plate collections. By detecting a period change with Yerkes plates and verifying with DASCH, a robust result can be achieved and the validity of glass plate analysis may be solidified.

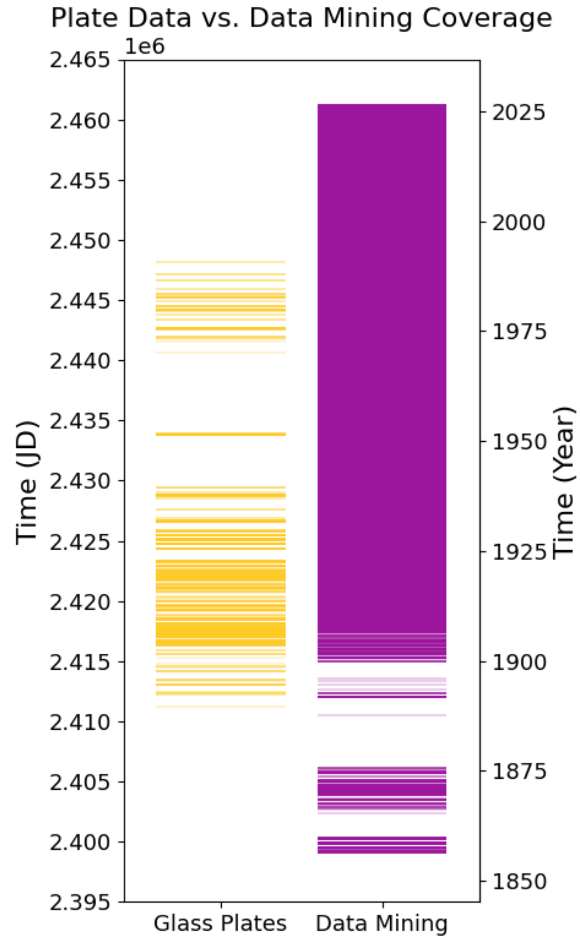

Figure 1: R Aql Event Plot of Yerkes Plates vs. AAVSO, Gaia, and ASAS-SN Data

1.3 Target Selection

Prior to beginning the project, the Gaia DR3 SOS variability catalogs were downloaded and cross-matched with *gaiadr3.gaia_source* with brightness, sky, and physical cuts. This resulted in a high-priority list of 579 Miras with $BP < 15$ and $G < 13$. The necessity of the double cut comes from the detection magnitudes of the glass plates. Due to the silver-halide emulsions of photographic plates, they are predominantly blue-sensitive, meaning the detection magnitude will be closer to Gaia BP than G. This is particularly important since Miras are primarily red, often leading to a dramatic BP-G gap and the possibility of a bright target in G being undetectable

Name	RA (deg)	Dec (deg)	BP (Mag)	$N_{25\%}$	Span (yr)	Current Period (d)	Visibility	Notes
R Aql	286.593	+08.229	9.59	393	95.1	270	Summer	Validation star, known CPC
KZ Her	282.801	+12.457	12.10	328	93.2	298	Summer	No prior period change studies
RW Vul	300.996	+12.705	11.54	187	88.2	206	Summer	Shortest period → most cycles
S Ori	82.254	−04.692	11.86	242	102.4	425	Winter	Longest baseline, known MPC
U Ori	88.955	+20.175	10.84	381	91.9	374	Winter	Second highest plate count
ST Sct	282.291	−12.928	13.0	404	62	219	Summer	Highest plate count

Table 1: Top targets for combining glass plate data, various databases, and new observations

in the glass plates.

The list of 579 Miras was then cross-matched with the Yerkes plate database to identify stars with over 100 plate epochs within the inner-25 percent field-of-view. Of the 93 Miras found, the top 5 (and 1 additional) were selected for analysis based on baseline, plate count, and decade coverage. See Table 1 for selection critera and a description of the top candidates in order of priority.

Note that ST Sct is not considered for the first run, as it has the lowest baseline, but it is the next target to consider if the first five are successfully completed. However, it will be of particular interest once the project is completed because AAVSO does not have any data on it, so even a 62 year baseline will be significant with 404 plates to analyze.

The highest priority star is R Aquilae, and it has become the main focus for the summer portion of the study. R Aql will serve as the validation star for the project, as it has been extensively studied to show a declining CPC. With a high plate count and visibility in the summer, it makes for the perfect target to compare modern and historic data at Yerkes. Coupled with the extensive baseline already provided by AAVSO and other organizations, we aim to recover the period change published in literature to validate our methodology so that we may move on to targets such as KZ Her that have undergone no previous period change studies.

2 Data Analysis

The data analysis has been split into three portions: data mining of online archives and databases, scanning and analyzing Yerkes Observatory glass plates, and taking and processing new observations using the Yerkes 24" reflecting telescope. Due to the lengthy process of plate scanning, R Aql has remained the primary target for analysis this summer.

2.1 Data Mining

The Amateur Astronomer Variable Star Organization (AAVSO), Gaia Date Release 3 (Gaia DR3), and the All Sky Automated Survery for SuperNovae (ASAS-SN) are the databases initially selected for data mining. Other databases such as APASS, AFOEV, and RASNZ-VSS may be included in future studies, particularly when it comes to stars with little data documented by AAVSO. After requesting and downloading light curve data from each database for each star on the target list, we discovered that AAVSO had very little data on KZ Her and RW Vul, with nothing on ST Sct. The later data for R Aql, S Ori, and U Ori also all show Henden bumps due to the addition of CCD observations and the updates of the comparison star databases. In the future, we plan to adjust and reduce the impacts of Henden bumps.

Once we had the data for each star, analysis of R Aql began. However, Gaia uses Gaia G, BP, and RP, which first needed to be converted to Johnson V, as that is what the majority of AAVSO data is in. The following equation from Table 5.7 of the Gaia Early Data Release 3 documentation was used for the conversion:

$$G - V = -0.02704 + 0.01424x - 0.2156x^2 + 0.01426x^3 \tag{1}$$

where $x = G_{BP} - G_{RP}$.

Additionally, ASAS-SN includes both V and g filters. We masked data with the g filter; however, this reduced the data from 2016 observations to 233. Moving forward, we plan to convert the g filter data to V for a more precise study.

We also needed to convert the Gaia time from TCB to JD. First we added the offset of 2455197.5 provided by Gaia to correct the time. Then, we converted TCB to JD using the astropy.time package. Now, with all of the data in the V band and JD time, we plotted the light curves separately and combined (See Figures 2 and 3).

With the combined light curve, we could begin the period analysis. Because the time-series data is uneven and awkwardly spaced, Weighted Wavelet Z-transform (WWZ) was used as advised by previous

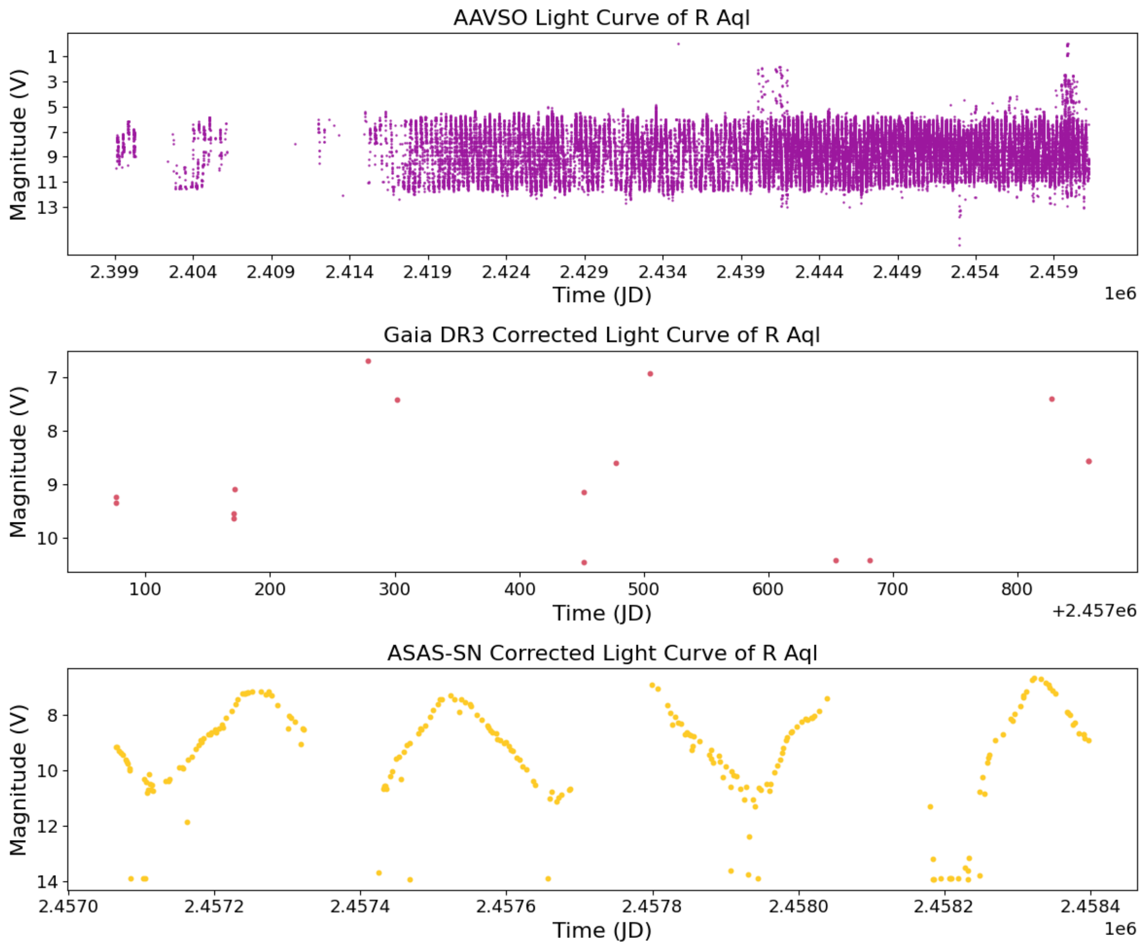

Figure 2: Light Curve from each database corrected for filter and time

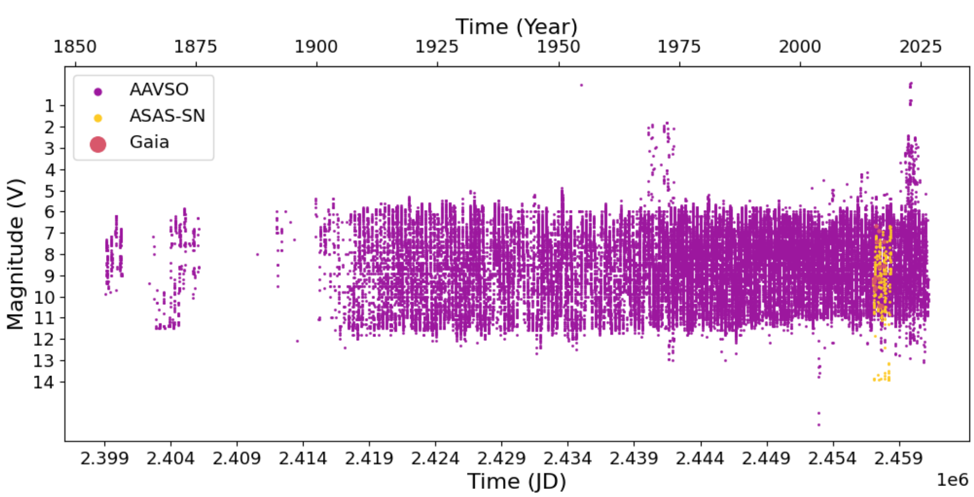

Figure 3: Light curve after combining AAVSO, Gaia, and ASAS-SN data

literature on Miras. WWZ was first developed by Grant Foster in 1961 and it is particularly beneficial when it comes to changing periods. This project used the python package libwwz from pycleolim as it is up to date with the most recent Python update.

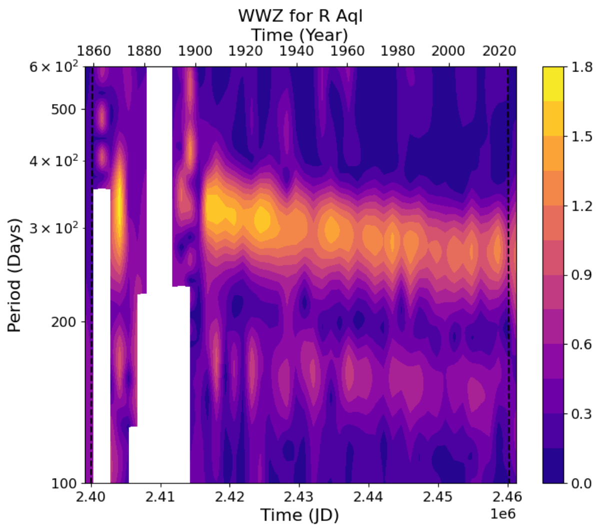

Figure 4: WWZ Plot showing R Aql's CPC

The WWZ results in a CPC decline that matches up with literature. Zijlstra and Bedding [5] found that the period for R Aql was around 350 days in the late 1850s and has slowly declined to around 270 days in the present day. As seen in the WWZ graph, we were able to recover this period decline from around 350 to 270 days. The gap in the graph is due to a lack of data during this time period, so we turn to glass plates to uncover a more precise period analysis.

2.2 Glass Plates

Despite the initial cross-match resulting in 393 plates with R Aql, not all 393 provide good data. Some had failing emulsion, others were too out of focus, and many did not truly include R Aql within the field of view. To find plates more efficiently, the Yerkes Astronomy Plate Searching System (YAPSS v3.0.0) was utilized to pinpoint plates with R Aql. YAPSS, initially developed by Dylan S. Caudill and updated by Emily Wade, works by using the digitized logbooks to find plates that depict certain areas of the sky.

Over the course of the project, 29 plates were scanned using a commercial Epson 10000 scanner. To begin analyzing plates, we first converted the scanned .tif files to .fits, providing header data from the finding aids and cleaning them by finding artifact pixels and masking them and their surroundings. Finally, the clean .fits file was uploaded to nova.astrometry.net. First, python was used to do so; however, after issues with the $force\_image\_upload$ command, files were uploaded directly to the website to retrieve WCS files. Of the 29, 5 were unable to be plate solved by nova.astrometry.net, due to either excessive writing on the plates, the use of a diffraction grate when exposing, or being a positive copy. Figure 5 shows a plate from the R series and its suspected reasons for failure.

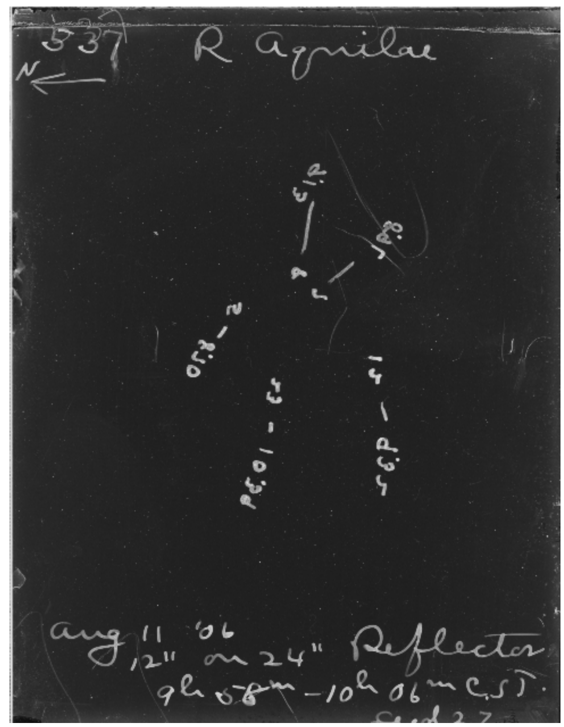

Figure 5: Full scan of R 537; likely failed due to excessive writing and scratches

Of the remainder of the plates, 12 will provide good photometric data. The oldest plate dates back to 1894 and was taken at the Lick Observatory. The subsequent plates were all taken at Yerkes Observatory, with the latest from 1922. From these, plate analysis continued, starting with testing various sigma values. Once the best sigma was determined, background detection began using different meshes and filter sizes chosen based on the RMS they gave back. The background was then subtracted, sources were detected, fluxes were measured, and an object table was created with SEP to be cross matched with a TOP 300,000 Gaia query. Though a match table was returned, the glass plate and Gaia have different magnitudes, so an additional step needed to be taken. Using the following equation, we considered how the Gaia magnitude

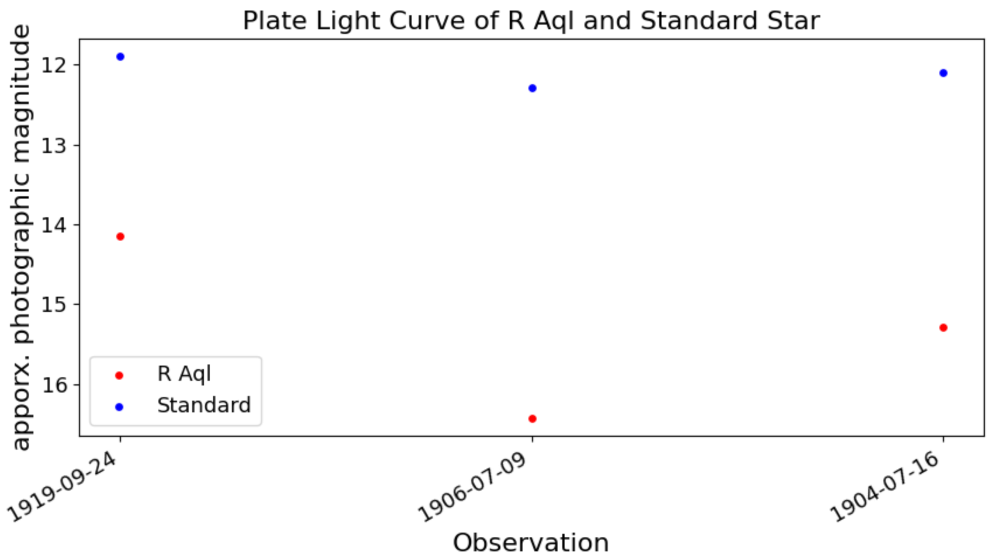

Figure 6: Light Curve from three fully analyzed plates

would appear if it were taken on a glass plate:

$$pg = phot\_bp\_mean\_mag + \alpha(bp\_rp) + ZP \qquad (2)$$

where pg represents the approximate photographic magnitude, $phot\_bp\_mean\_mag$ and bp_rp come from Gaia, α comes from the color calibration, and ZP is the zero-point magnitude.

At this point in the project, only three plates have been fully analyzed to reveal a pg magnitude for R Aql. Moving forward, these magnitudes will be added to the light curve to improve the precision of the WWZ period analysis. However, I am still skeptical as to how accurate the results have been. Currently, the plates are returning bp magnitudes around 14-16 magnitude (see figure 6). When compared to a standard star nearby, R Aql does vary more, but when compared to the bp magnitude of R Aql from Gaia, it is dimmer by around 6-8 magnitudes. Before continuing analysis of other plates, I plan to re assess the code to determine if there are any issues. More plates from Yerkes to bridge the gap will eventually be included, but I would also like to look into other glass plate databases such as DASCH to see if I can fill in the curve from 1880 to 1890.

2.3 Current Observations

Aside from the historic Great Refractor, Yerkes also maintains two modern research telescopes, the 24-inch reflector and the 40-inch reflector, which were installed in the 1960s. Throughout the summer, various observations of R Aql and the second target star, KZ Herculis, were taken. Although back to back observations of Miras are not necessary due to their long period, it is an effective way to demonstrate both the historic and modern significance of Yerkes in research.

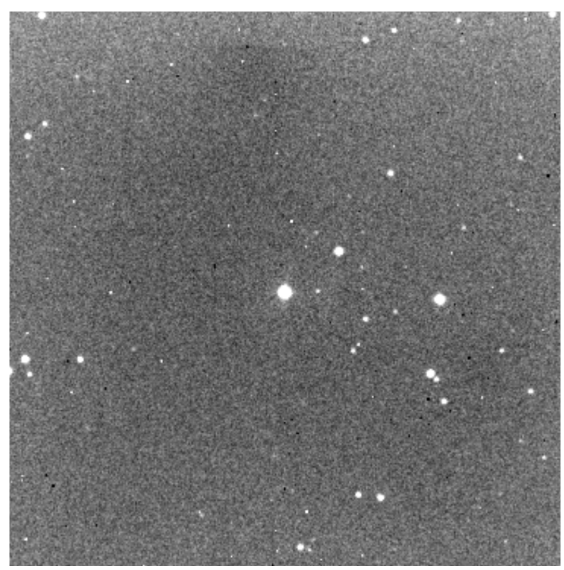

Figure 7: R Aql centered in 60s exposure in B band

From the observations, we can learn more about the conversation between CCD observations taken today and photographic glass plates taken nearly 100 years ago. In total, 8 observations were taken of R Aql from

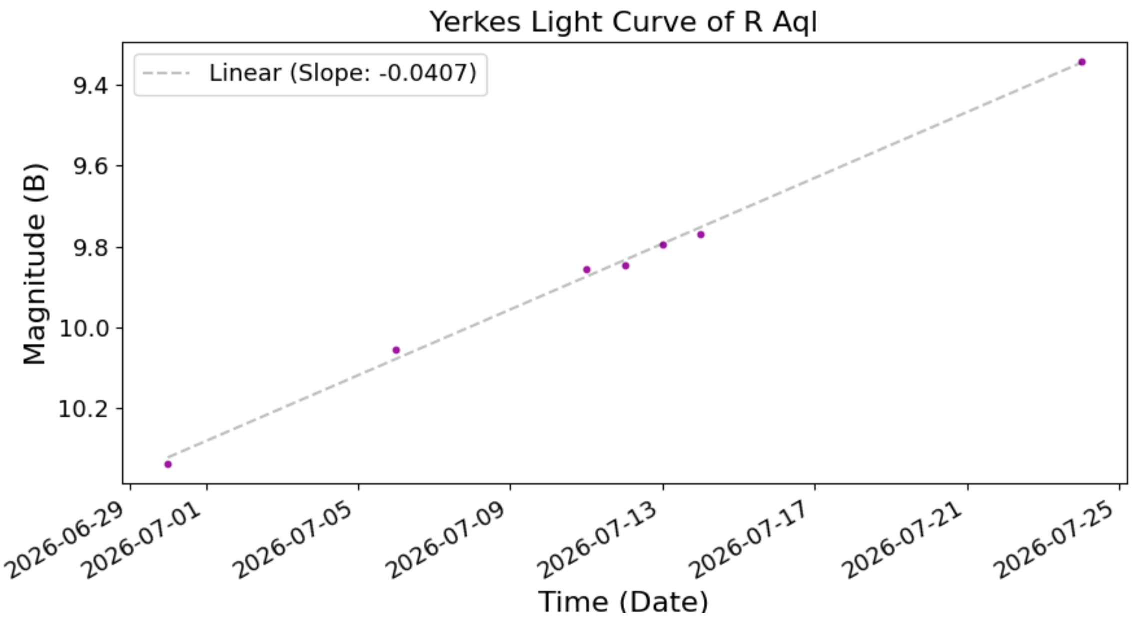

Figure 8: Light Curve of R Aql from 24-in Reflector calibrated with single star with curve fit

June 30th, 2026 to July 28th, 2026. For KZ Her, 4 observations were taken from July 12th, 2026 to July 24th, 2026. Figure 7 shows a photo of R Aql taken through the 24-inch reflector.

Each observation was calibrated using master flats, darks, and biases and then analyzed with aperture photometry. For R Aql, the star with AUID 000-BCF-220 was used as the comparison star, chosen because it is not a variable star and has knwon values for V and B-V. In Figure 7, it is down and to the right of R Aql, making up the bottom most corner of the bright trapezoid.

To perform photometry, we attempted two methods. The first was to calibrate against a single standard star in the frame. The target star profile was visualized and fit with a gaussian curve based on the test aperture and the annulus. Then the pixel values in each aperture were summed for the target and comparison star. We averaged the sky counts from the sky annuli to subtract the sky counts from the star aperture counts. To correct for observing conditions, we divided the counts of the target star by the counts of the comparison star. Once this process was completed for each science frame, we began photometry and looped through various aperture sizes. For each aperture, a normalized light curve was plotted and the percentage of point-to-point scatter was calculated. When plotting the optimal light curve, the FWHM throughout the observing run was also considered.

With the optimal aperture size for the target was determined, we began photometry for the comparison star. The process was repeated and the optimal aperture for the comparison was decided. Now, we moved on to magnitude calculation, starting with finding the Zero-Point flux from the comparison star. First, we calculated the B magnitude of the comparison using Equation 2 and the V and B-V values from AAVSO's Variable Star Plotter Photometry Chart.

$$B = V + (B - V) \tag{3}$$

Then, we input the magnitude into the zero-point flux equation:

$$ZP = B_{comp} + 2.5\log_{10}(\frac{counts_{comp}}{exposure}) \tag{4}$$

This was completed for each image in each observation, and then applied to calculate the magnitude for the target star using Equation 4.

$$B_{target} = ZP - 2.5\log_{10}(\frac{counts_{target}}{exposure}) \tag{5}$$

For each observation, the B magnitude was averaged and plotted to create a light curve, as seen in Figure 8.

To ensure our magnitude calculations were accurate, we cross-checked with the expected brightening rate over the time period. Because R Aql does not brighten at the same rate it dims, we used only 0.4 of

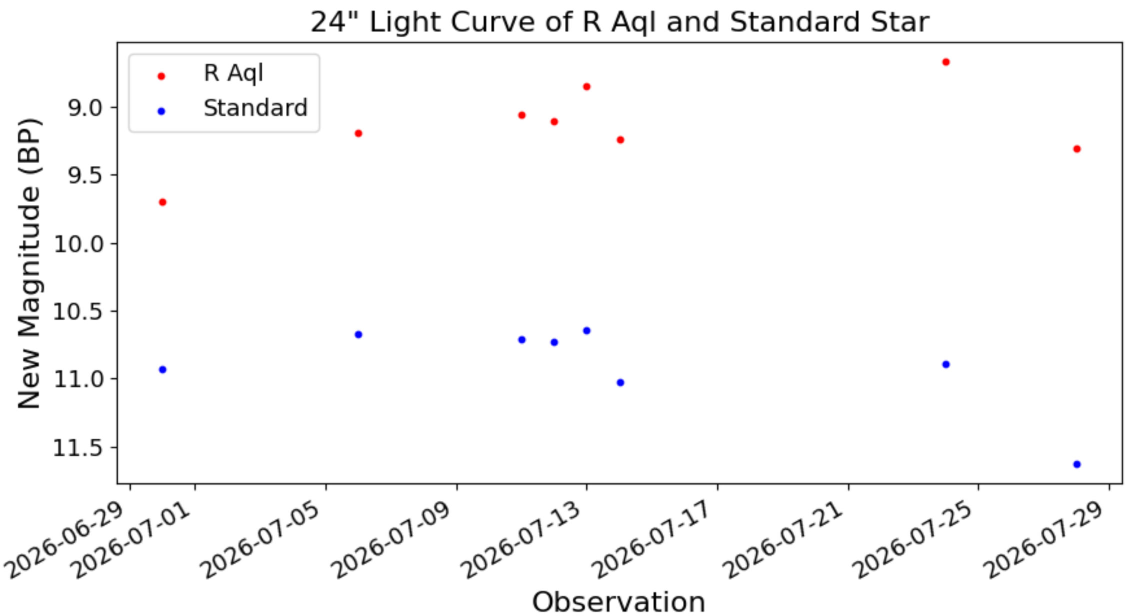

Figure 9: Light Curve of R Aql from 24-in Reflector calibrated with Gaia

the period to account for the difference. Additionally, the change in magnitude was taken to be 6.5 magnitudes, as R Aql's magnitude ranges from about 5.5 to 12 magnitudes.

$$\frac{M_{max} - M_{min}}{time} = \frac{6.5 \ \ mag}{0.4 * 270 \ \ days} = 0.0602 \ \ mag/day \tag{6}$$

When compared to our rate of change, the expected is quicker than our measured of 0.0407 mag/day. This difference may be accounted for by the small sample size and recent issues with the 24-in reflector's camera power supply.

The second method was to calibrate against the Gaia catalog, as this should be more accurate. This is a similar process as that used for the glass plate photometry. For each night of observations, we started by running a $TOP \ \ 300,000$ Gaia query for the frame. Then, we did source detection using DAO Star Finder after detecting and subtracting the background. To determine the proper radius to perform aperture photometry, we started by fitting a Gaussian to the data. Then, we found where the second derivative of the Gaussian reached zero, indicating an accurate radius. From this radius, we obtained a circular aperture and sky annulus. The pixel values from both aperture sizes were summed and averaged so that the expected contribution from the sky could be subtracted from the star aperture. The x and y centroids of the star were matched to RA and Dec values, which were then compared to Gaia coordinates from the query to return a table of matching stars and their Gaia magnitudes.

Now, with the match table, we could begin calculating magnitudes. The first step was to find the zero-point flux again. This time, we used Equation 7 for each matched star and then averaged the list of zero-points to use for further calculations.

$$ZP = phot\_bp\_mean\_mag + 2.5\log_{10}(\frac{flux}{exposure}) \tag{7}$$

Next, we used the ZP to find the magnitude:

$$new \ \ mag = -2.5\log_{10}(\frac{flux}{exposure}) + ZP \tag{8}$$

We repeated these steps for the standard star BD+08 3968. For each night of observations, we cleaned the list of magnitudes by dropping any Nans and filtering out bad detections that were skewing the data. Then we averaged them and plotted the average of each night for both R Aql and the standard star (see figure 9).

As I was not convinced with the glass plate magnitude calculations, I am unhappy with the results of this calibration so far. I am skeptial of the pattern both R Aql and the standard star seem to follow, and I am considering dropping the observation from the night of July 28th, 2026 because it was hazier than anticipated, which could be dragging the magnitude down.

Moving forward, there will be difficulties combining these observations with the data mined light curve because they are in B magnitude rather than V. B magnitude was chosen because it most directly matches the effects from the emulsion on the glass plates. BP was used for the Gaia calibration because it is closest to B. Rather than estimate the B-V color index to convert from B to V, instead, I suggest continuing observations with the 24-inch reflector to get a full cycle and finding the period without the data mined observations. In this way, we can directly compare modern observations taken at Yerkes with historic glass plates and recover the period decline using the plate epoch and modern epoch only.

3 Results

Until I reassess the magnitude calculations from the plates, the use of Yerkes plates in the study of Mira variables has not yet been validated. As of now, the most successful part has been through data mining, where I was able to recover the Continuous Period Decline of R Aql published in literature. I have also completed the photometry on my own observations of R Aql and recovered a similar rate of change in magnitude through one method, but not through the more accurate method. Photometry for the KZ Her observations is underway, but in the mean time, I have created notebooks and an outline to follow for future studies of Miras using glass plates, AAVSO, Gaia, ASAS-SN, and Yerkes 24-in reflector observations.

4 Discussion

As magnitudes have yet to be confirmed, the discussion will be targeted more towards next steps.

4.1 Validation

To validate the use of Yerkes glass plates in the study of period-changing Mira variables, we will need to complete the analysis of the remaining glass plates already scanned by performing further photometry. I am not yet convinced that my current work with the glass plates has been accurate, due to how dim R Aql is appearing. Moving forward, we will need to scan a larger set of plates around a single time period to perform an individual WWZ to recover the theorized period at that time before adding the calculations to the larger light curve.

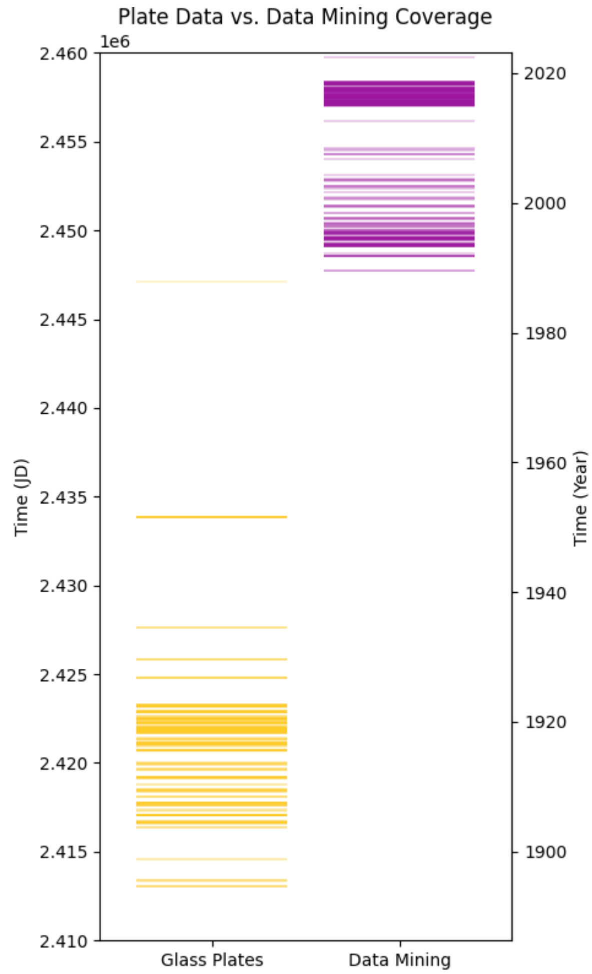

Figure 10: KZ Her Event Plot of Yerkes Plates vs. AAVSO, Gaia, and ASAS-SN Data

4.2 Application

Once the methodology has been validated, we can move on to targets that have more interesting cases, with KZ Her and ST SCT being of most interest.

KZ Herculis will be the next highest priority, as no prior period change studies have been completed on it. AAVSO's first observation is around 1990, meaning that Yerkes glass plates could reveal an unknown period change after extending the baseline by nearly 100 years (see Figure 10).

ST Scutis, though initially low priority, has moved up the list because AAVSO has no data on it despite Yerkes having the most plates including it. This could be an opportunity to extensively study a Mira that has very little data anywhere but here.

5 Conclusion

We conclude that further work is needed, particularly in the photometry of glass plates and 24-inch reflector observations. We started by splitting data analysis into three portions: data mining, glass plates, and our own observations. By doing so, we were able to obtain magnitudes of R Aql from various time periods, but the validity of calculations for the latter two portions is not yet confirmed. For data mining, we could successfully recover the Continous Period Declined experienced by R Aql. For glass plates, we determined that the intial cross-matching results were not representative of how many plates would actually provide data, and for our own observations, we have not yet determined the most accurate way to perform photometry. Moving forward, we would like to improve upon the code for the photometry of glass plates and observations before continuing on the the next target, KZ Herculis.

5.1 Acknowledgements

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